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AffectLab: Design and Evaluation of an Emotion-Aware Multimodal System for Adaptive Difficult-Conversation Rehearsal

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Abstract

Practising difficult interpersonal conversations can be inaccessible, inconsistent, or uncomfortable, while conventional conversational systems often ignore context, vocal cues, and uncertainty. This dissertation presents AffectLab, a non-clinical application for adaptive communication rehearsal that combines contextual text and speech emotion recognition with deterministic dialogue, feedback, safety, and fallback policies. Emotion models were evaluated on five speaker-independent IEMOCAP folds using classification, calibration, and dialogue-level bootstrap measures; the integrated application was assessed through authored cases, automated and full-stack tests, accessibility checks, restart recovery, and local latency measurements. Three-turn context increased text macro F1 from 0.6651 to 0.7332. Equal-weight text–audio fusion reached 0.7883 macro F1 and 0.7832 accuracy, improving macro F1 over contextual text by 0.0551 (95% interval 0.0407–0.0713), although its ECE of 0.1619 exposed a calibration trade-off. The application passed 245 backend tests, 76 desktop/mobile browser cases, production frontend checks, and two database-backed restart journeys; deterministic conversation turns had a 5.836 ms 95th-percentile local latency. In a supplementary single-arm pilot, 12 of 15 enrollees completed the protocol, while realism and feedback usefulness each had a median rating of 5/7. Required pilot tasks bypassed multimodal inference and affect-aware pacing, so these ratings describe only the baseline rehearsal workflow and provide no evidence about the acceptability or benefit of emotion-aware adaptation. The small, uncontrolled pilot also cannot establish causal benefit or real-world transfer. Within these limits, AffectLab demonstrates an implementation-complete, reproducibly evaluated, uncertainty-aware system whose suitability for controlled deployment depends on target-environment validation.

Keywords: affective computing; emotion recognition; multimodal fusion; conversational context; difficult-conversation rehearsal; human–computer interaction.

Statement of Scientific integrity

I declare that this MSc thesis is an original report and represents my own work, conducted under supervision of Lect. Dr. Todor Ivașcu. All research methods, data collection, analysis and interpretation presented in this thesis are accurately reported and all sources have been cited appropriately where I have made use of ideas and/or visuals of others.

I used generative AI during the writing and research process. The details are in the Appendix A.

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List of Abbreviations

AI	Artificial intelligence
API	Application programming interface
BERT	Bidirectional Encoder Representations from Transformers
BSON	Binary JSON
CI	Continuous integration
CORS	Cross-origin resource sharing
CPU	Central processing unit
CSV	Comma-separated values
DOM	Document Object Model
ECE	Expected calibration error
ERC	Emotion recognition in conversation
F1	Harmonic mean of precision and recall
GDPR	General Data Protection Regulation
GPU	Graphics processing unit
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IEMOCAP	Interactive Emotional Dyadic Motion Capture
JSON	JavaScript Object Notation
JWT	JSON Web Token
LLM	Large language model
NLL	Negative log-likelihood
P95	95th percentile

PCM	Pulse-code modulation
RQ	Research question
SD	Standard deviation
SER	Speech emotion recognition
SHA-256	Secure Hash Algorithm 256-bit
SMTP	Simple Mail Transfer Protocol
TLS	Transport Layer Security
TTL	Time to live
UA	Unweighted accuracy
UI	User interface
WA	Weighted accuracy
WAV	Waveform Audio File Format
WCAG	Web Content Accessibility Guidelines

Chapter 1

Introduction

1.1 Context and Problem

Difficult interpersonal conversations arise in ordinary situations such as negotiating workload, maintaining a personal boundary, or expressing a relationship need. Although a person may know what they want to communicate, pressure and disagreement can make it harder to state a clear request, respond constructively, or reflect on alternative wording. Structured role-play offers a way to practise these skills before a real interaction, particularly when it provides clear objectives, resistance, reflection, and feedback.

Conventional practice may require another person, a facilitator, or a scheduled and potentially uncomfortable session. A digital system can instead provide private, repeatable access and preserve the same scenario for comparison between attempts. Language models can add conversational variety, but unrestricted generation is not a sufficient foundation for dependable practice: a model may leave its assigned role, vary the intended difficulty, or produce feedback unsupported by the user's words. Scenario progression, outcomes, assessment, and safety therefore require explicit control.

A second challenge concerns affect. Meaning can depend on preceding dialogue and on acoustic delivery as well as the current transcript, which motivates contextual and multimodal emotion recognition. Yet a benchmark label is not a direct measurement of a person's internal state. Predictions can be wrong, overconfident, or inconsistent across text and audio, particularly outside the data on which the models were evaluated. An application must therefore decide not only how to infer affective information, but when that information is sufficiently reliable and appropriate to influence an interaction.

This dissertation addresses the integration problem created by these requirements: how to combine contextual text and speech models with flexible role-play while keeping dialogue outcomes, feedback, safety, privacy, and failure handling predictable. AffectLab treats affect predictions as bounded evidence for modest and reversible pacing choices, not as authoritative psychological judgements. Deterministic policies remain responsible for scenario state, observable communication features, and safety precedence, while optional generative services may realise an already constrained response. Model performance, application conformance, and supporting user feedback are evaluated separately so that evidence from one level is not used to justify claims at another.

1.2 Aim and Objectives

The aim of this dissertation is to design, implement, and evaluate AffectLab, an emotion-aware system for rehearsing difficult interpersonal conversations. The system combines structured role-play, contextual text and audio-based affect recognition, uncertainty-aware adaptation, and evidence-grounded feedback while preserving transparency, user control, privacy, safety, and deterministic fallback behaviour.

The objectives are:

1. to establish, from relevant research, the design constraints for structured communication rehearsal, contextual and multimodal affect recognition, calibrated uncertainty, and responsible human–AI interaction;
2. to develop and compare utterance-only text, contextual text, audio, and late-fusion models under a speaker-independent protocol, evaluating classification and probability quality as distinct concerns;
3. to design and implement scenario-specific dialogue policies and a bounded adaptation policy in which model confidence, modality agreement, availability, user preference, and safety determine only permitted, inspectable actions;
4. to provide evidence-linked feedback and reflective functions within a privacy- and safety-conscious application whose core workflow survives optional-service failure; and
5. to evaluate model behaviour and integrated application quality with evidence appropriate to each claim, while reporting the small user pilot only within its non-adaptive scope.

1.3 Research Questions

The dissertation is guided by four research questions:

RQ1. Does incorporating preceding conversational turns improve text-based emotion recognition compared with classifying each utterance independently?

RQ2. Does combining contextual textual information with acoustic information improve emotion-classification performance compared with the individual text and audio modalities, and what trade-off exists between classification performance and confidence calibration?

RQ3. How can uncertain multimodal affect predictions be integrated into a structured role-play dialogue policy while keeping the system's behaviour explainable, controllable, safe, and robust to disagreement or inference failure?

RQ4. To what extent does the completed AffectLab application satisfy its functional, reliability, recovery, accessibility, responsiveness, and deployment requirements under reproducible engineering evaluation?

1.4 Approach and Contributions

AffectLab uses a layered architecture that separates affect inference, dialogue control, response realisation, feedback, and safety. The experimental pipeline compares an utterance-only text classifier with a model that adds preceding causal dialogue turns. A separate speech model provides acoustic evidence, and late fusion combines the modality distributions. Classification metrics establish predictive performance, while calibration measures assess whether confidence is suitable for operational thresholds. Speaker-independent session folds and paired dialogue-level comparisons keep the principal model comparisons aligned.

Within the application, a deterministic controller represents each scenario as stages, character behaviour, completion conditions, and unresolved outcomes. It selects the next conversational action from observable evidence rather than allowing a language model to decide whether the user has succeeded. An optional generative service may phrase that action more naturally, but validation and deterministic templates preserve the recorded decision when generation is unavailable or unsuitable.

Multimodal estimates enter the interaction through a restricted affect policy. Confidence, agreement between modalities, inference availability, eligibility, and an explicit user preference determine whether the system preserves baseline behaviour, offers a pacing choice, or permits a brief acknowledgement. Safety interruption takes precedence and does not depend on the emotion classifier. This separation also prevents affect predictions from determining scenario success, feedback scores, or participation decisions.

Feedback is derived from inspectable language features associated with each scenario, such as a clear request, maintained boundary, non-blaming statement, acknowledgement, or practical next step. The relevant conversational turns remain available as evidence. Replay, retry, branching, comparison, personal takeaways, and action cards support reflection on alternative responses rather than reducing practice to a single score.

Evaluation follows the same separation of responsibilities. The model study addresses context, modality complementarity, and calibration. Authored policy cases and automated checks address dialogue transitions, fallbacks, persistence, recovery, accessibility, responsive behaviour, and the complete application workflow. Local timing measurements characterise only the tested deterministic path. A small pilot supplies supplementary feedback on the baseline rehearsal workflow; because its required tasks did not exercise multimodal adaptation, it is not evidence that adaptation benefits users.

The dissertation makes five principal contributions:

1. **A controlled evaluation of causal conversational context and multimodal complementarity.** Utterance-only text, contextual text, speech, and fused configurations are compared on matched speaker-independent data, with dialogue-level uncertainty estimates.
2. **An explicit separation between classification performance and operational confidence.** Predeclared fusion and validation-fitted calibration expose the trade-off between the strongest classifier and the probability estimates used by an interactive policy.

3. **A bounded architecture for affect-aware role-play.** Deterministic dialogue and safety logic remain authoritative, while uncertain affect estimates and optional generation are limited to inspectable, reversible actions with defined fallbacks.
4. **Evidence-grounded rehearsal and reflection.** Scenario-specific outcomes and feedback are connected to observable turns, and replay and branching allow alternative responses to be compared from the same conversational context.
5. **A layered verification framework for an integrated application.** Model evaluation, policy conformance, application verification, local performance, and supplementary user feedback are kept distinct, making the scope of each resulting claim explicit.

Together, these contributions concern the responsible integration of affective models into an implemented communication-practice system. They do not imply that AffectLab identifies a user's true emotion or that completing a rehearsal improves a real relationship.

1.5 Scope and Dissertation Structure

AffectLab is intended for adults rehearsing ordinary difficult conversations. The implemented scenarios cover workload negotiation, personal boundary-setting, and expression of a relationship need. Text interaction is primary and voice input is optional. The affective component is limited to supervised categorical recognition from text and audio; it excludes video, physiological signals, multilingual inference, diagnosis, therapy, autonomous crisis assessment, and consequential decisions about another person. Emotion estimates are uncertain application inputs, not findings about an individual's internal state.

The empirical model claims are limited to the defined IEMOCAP protocol. Its ten actors, controlled English-language interactions, annotation process, and categorical labels do not establish generalisation across cultures, accents, devices, populations, or spontaneous application use [BBL+08]. Deterministic language rules are reproducible but cannot recognise every appropriate paraphrase and are not validated measures of general communication ability or psychological constructs. Optional providers introduce additional privacy and reliability boundaries, so the text-based fallback remains part of the intended system rather than an exceptional mode.

Application evaluation establishes behaviour only for the tested cases and configurations. It does not replace independent security review, target-environment load and recovery testing, live-provider evaluation, participatory accessibility work, or validation of trained inference under deployment conditions. The small, uncontrolled pilot concerns the non-adaptive baseline workflow and cannot establish causal benefit, sustained skill development, or transfer to real conversations. AffectLab is therefore presented as an implementation-complete, deployment-oriented research application whose wider use remains conditional on further technical, ethical, and human evaluation.

Chapters 2–6 cover the literature, architecture, implementation, evaluation, and conclusions, respectively.

Chapter 2

State of the Art

Emotion-aware communication rehearsal combines several research traditions whose evidence should not be conflated. Communication-training studies can establish the value of structured practice, conversational-agent research can inform interaction design, and affective-computing benchmarks can measure agreement with operational emotion labels. None of these alone demonstrates that an integrated application is educationally effective, reliable for a new speaker, or safe. This chapter therefore reviews only the literature needed to motivate the research questions and design choices. It distinguishes throughout between predicting a dataset label, controlling application behaviour, and improving a person's communication skills.

2.1 Communication Rehearsal, Role-Play, and Feedback

Role-play turns interpersonal knowledge into observable decisions: a learner must formulate a request, respond to resistance, clarify an intention, and decide whether to maintain or revise a position. Compared with a consequential real interaction, a simulation permits mistakes, repetition, and reflection. Its value nevertheless depends on its learning objectives and feedback, not simply on conversational realism. In a randomised study of 103 medical students, Bosse et al. found that peer role-play and standardised-patient training improved communication performance and self-efficacy relative to a control condition [BSN+12]. A systematic review of 14 virtual-patient studies similarly associated stronger interventions with preparation, scaffolding, debriefing, reflection, and feedback, while finding heterogeneous evidence for comparative effectiveness [LKK+20]. These clinical results support structured rehearsal but do not establish equivalent transfer to everyday workplace or relationship conversations.

Language models can broaden the range of situations available for rehearsal. IMBUE combines personalised interpersonal simulations with just-in-time, framework-based feedback; its randomised evaluation with 86 participants reported gains in measures including self-efficacy and skill mastery, but transfer to new and more difficult situations was more limited than improvement within trained situations [LSR+24]. The result is important because it treats an LLM interaction as a learning activity rather than an open-ended chatbot. It also shows why performance inside a familiar scenario, perceived realism, immediate confidence, and transfer are

separate outcomes.

Feedback is most actionable when it relates a learning goal to observable performance and a feasible next step [HT07]. In communication rehearsal, evidence might include whether the learner made a specific request, gave a reason, acknowledged another perspective, or maintained a stated boundary. Automated evidence remains a proxy: a lexical or rule-based detector can miss a valid paraphrase or misread quotation, negation, and sarcasm, whereas a fluent generated explanation can sound authoritative without being valid. Traceable feedback should therefore identify the turn that triggered an assessment and permit review, retry, or comparison of alternative continuations. Such mechanisms support reflection, but their presence does not prove improvement outside the system. This literature motivates defined scenario goals, legitimate unresolved outcomes, repeatable attempts, and evidence-linked feedback, while requiring any learning claim to be evaluated independently.

Repeatability creates an additional evaluation risk: learners may discover the phrases rewarded by one scenario without acquiring a strategy that survives changed wording, resistance, or context. Evidence recorded inside the simulator should therefore be distinguished from independent assessment and from transfer to an unfamiliar conversation. Self-reported confidence can describe a participant's immediate experience, but without a suitable comparator it cannot identify whether practice, feedback, novelty, or expectation produced the change. This distinction is particularly important for short pilots, which are useful for finding workflow problems and estimating acceptability but rarely support durable learning claims.

2.2 Deterministic and Generative Conversational Agents

Classical task-oriented dialogue systems make conversational state and action selection explicit. RavenClaw, for example, separates domain-specific task logic from reusable dialogue-management capabilities such as turn-taking and recovery [BR09]. For rehearsal, explicit states can represent raising an issue, clarifying a concern, responding to resistance, and reaching either a completed or unresolved outcome. Their principal benefit is controllability: transitions and permissible endings can be explained and tested. Their costs are policy authoring, incomplete coverage of natural phrasing, and repetitive surface responses.

Large language models provide more flexible linguistic realisation and can respond to inputs not anticipated in a template. Research on LLM role-play shows how personas, dialogue goals, and recovery strategies can structure multi-turn simulation [TSG25]. Flexibility is not the same as control, however. In a comparison of generated and human-authored professional role-play responses, Lu et al. found that judged quality deteriorated across turns in the evaluated setting, particularly for naturalness and context maintenance [LJG25]. A plausible reply can also damage a rehearsal exercise by dropping a scenario constraint, becoming prematurely agreeable, or declaring success before its conditions have been met.

A hybrid therefore offers a defensible division of responsibility: deterministic policy selects the stage, resistance level, completion status, and allowed response plan; optional generation varies wording within that plan; and a deterministic re-

sponse remains available if generation fails. Related work on emotionally conditioned and empathetic generation shows that affect-related signals can influence perceived response quality [ZHZ+18, RSLB19], but perceived empathy neither verifies the inferred emotion nor establishes an appropriate instructional action. In a practice scenario, scepticism or refusal may be intentional. Affect-related adaptation should consequently remain subordinate to scenario goals, user choice, and independent safety rules.

2.3 Contextual Text and Speech Emotion Recognition

Emotion recognition requires an operational target. Categorical systems assign labels such as anger, happiness, sadness, and neutral, whereas dimensional accounts describe affect using quantities such as valence and arousal [Ekm92, Rus80]. These representations are useful abstractions rather than direct measurements of private experience. Emotion, mood, feeling, expressive behaviour, and an annotator's interpretation are not interchangeable constructs [Sch05]; observable cues also vary with speaker, culture, setting, and communicative purpose. Accordingly, "emotion recognition" here means prediction of annotated expressive categories. It does not license diagnosis, inference of intent, or a claim that the system knows what a user feels.

The Interactive Emotional Dyadic Motion Capture database (IEMOCAP) provides about twelve hours of scripted and improvised dyadic interactions from ten actors in five sessions, with speech, transcripts, other modalities, and emotion annotations [BBL+08]. Its paired modalities and conversational form make it suitable for controlled text–audio comparison. A common four-class task uses anger, happiness (often including excitement), sadness, and neutral, although published work differs in label mapping, preprocessing, splits, and metrics [LWC+22]. Other conversational corpora, including the multi-party television dialogue in MELD and the written exchanges in DailyDialog, provide useful but non-equivalent domains [PHM+19, LSS+17]. A catalogue of datasets adds little to the present argument: the important point is that results are comparable only under compatible protocols, and an acted English corpus with ten speakers cannot establish robustness across populations, devices, languages, or spontaneous deployment speech.

Conversational samples also violate the assumptions behind an utterance-level random split. Adjacent turns share speakers, recording conditions, subject matter, and affective trajectories. Holding out an entire IEMOCAP session prevents the same actors from appearing in training and test data, while a separate session for validation can protect model selection from test leakage. It does not remove the shared studio, task, and annotation conventions. Statistical comparisons should also respect dependence within dialogues rather than treating every utterance as an independent participant.

Protocol comparability extends beyond the split. Merging excitement with happiness, discarding unmatched utterances, balancing classes, or reporting weighted rather than unweighted measures can materially change the apparent result. A strong published score is therefore context for method selection, not a target that can be compared without reconstructing its data definition. The dissertation in-

stead fixes one primary task and uses identical held-out items for the comparisons that answer its research questions.

Pre-trained text encoders such as BERT and RoBERTa provide contextual token representations that can be adapted to classification [DCLT19, LOG+19]. Yet an utterance such as "That's fine" remains ambiguous without its conversational setting. Emotion-recognition-in-conversation models address this limitation through speaker states, dialogue graphs, commonsense features, or representations of a speaker's previous utterances [MPH+19, GMP+19, GMG+20, LL22]. These studies motivate the use of dialogue history, but many offline arrangements are unsuitable for live interaction. A deployable classifier for turn t may use the current utterance and a bounded history,

$$\hat{y}_t = f(u_t, u_{t-1}, \ldots, u_{t-k}),$$

but not future turns or neighbouring gold labels. A bidirectional token encoder can still be used inside this causally assembled input; bidirectionality over its tokens does not imply access to the future conversation. A matched comparison of utterance-only and causal-context models consequently answers a clearer question than an uncontrolled comparison among unrelated architectures.

Speech contributes pitch, intensity, timing, rate, voice quality, and pauses that a transcript omits, but these cues also reflect identity, health, language, microphone, and environment. Compact descriptors such as GeMAPS offer interpretable acoustic baselines [ESS+16]. Self-supervised encoders including wav2vec 2.0, HuBERT, and WavLM instead learn representations from large speech collections [BZMA20, HBT+21, CWC+22]. Their success in speech representation learning is not itself evidence of emotion-classification accuracy, and a downstream model can still exploit speaker or recording artefacts. Speaker-independent evaluation is therefore essential. At application level, audio must also be optional and recoverable because permissions, noise, silence, short recordings, and the identifying content of a voice introduce failure and privacy costs absent from transcript-only inference.

2.4 Multimodal Fusion, Calibration, and Uncertainty

Text and speech provide complementary but differently structured evidence. Multimodal research distinguishes representation, alignment, and fusion as separate challenges [BAM19]. Early and cross-modal methods can model fine-grained interactions; for example, the Multimodal Transformer uses directional attention between unaligned streams [TBL+19]. Such capacity also increases coupling and makes missing inputs and decision provenance harder to handle. For the present research questions, decision-level fusion supplies a more transparent test. If $p_T(y)$ and $p_A(y)$ are independently produced text and audio distributions, their mixture is

$$p_F(y) = \lambda p_T(y) + (1 - \lambda) p_A(y), \qquad 0 \leq \lambda \leq 1.$$

Equal weighting predeclares $\lambda = 0.5$; a fitted alternative must estimate it using validation data rather than the final test set. The rule is modular, exposes modality disagreement, and permits text-only fallback. It is less expressive than cross-modal attention and can still become confidently wrong when both branches exploit misleading cues.

Classification quality and probability quality answer different questions. Accuracy and macro F1 assess predicted labels, while calibration asks whether confidence corresponds to observed correctness under the evaluated distribution. Neural classifiers are often miscalibrated; temperature scaling can adjust logit magnitudes using a held-out validation set without changing the usual model's top-class ordering [GPSW17]. More flexible multiclass methods such as Dirichlet calibration are possible, but require sufficient validation evidence and can overfit [KPNK+19]. Expected calibration error offers a useful summary but is bin-dependent, so it should be interpreted beside proper scoring measures or reliability evidence rather than as a complete guarantee.

Fusion makes this separation particularly important: the configuration with the strongest classification score need not have the best calibrated confidence. Training, weight selection, calibration fitting, and testing must therefore remain separated inside each speaker-held-out fold. Moreover, calibration is distribution-dependent and can degrade under shift [OFR+19]. A confidence threshold derived from IEMOCAP is an operational presentation rule, not a psychological boundary. Modality agreement is likewise informative but not proof. These limitations support restrained application behaviour: uncertain, conflicting, or unavailable evidence should lead to baseline dialogue or an explicit pacing choice, not to an imposed emotional interpretation.

2.5 Research Gap and Design Implications for AffectLab

Prior work establishes the components of the problem but not their complete evidential chain. Communication-training research supports structured practice, reflection, and feedback; contextual and multimodal models can improve benchmark label prediction; calibration research qualifies the meaning of confidence; and generative agents provide flexible dialogue with known multi-turn control risks. What remains is an integration-and-evaluation gap: a bounded rehearsal system must connect these components without treating benchmark accuracy as emotion understanding, fluent generation as policy compliance, application tests as educational validation, or immediate user ratings as evidence of transfer.

This gap yields four design implications. First, model comparisons require a common, speaker-independent protocol. The contextual text model should be compared with an utterance-only counterpart on the same held-out dialogues, and a separately trained speech branch and transparent fusion rule should be evaluated on matched text–audio cases. Classification and calibration must be reported separately, with test data excluded from model and policy selection. These choices provide evidence for the first two research questions while keeping conclusions specific to the benchmark.

Second, affect inference should be advisory rather than authoritative. Its permitted effects should be few, reversible, and inspectable; scenario progression, completion, feedback, and safety remain governed independently. This boundary is especially important because emotionally expressive input can occur during serious distress. Risks identified for generative wellness applications justify clear scope limits and a safety route that does not depend on a four-class emotion model [DFC24].

In the European legal context, the Artificial Intelligence Act also restricts particular emotion-inference uses in workplace and educational settings, subject to its stated scope and exceptions [Eur24]. Voluntary personal rehearsal is not equivalent to institutional surveillance, but deployment would still require case-specific legal, privacy, and ethical assessment. Explicit microphone activation, minimised audio retention, access control, and withdrawal procedures follow from this boundary rather than from a claim of compliance by design alone.

Third, hybrid dialogue should keep control decisions separate from surface wording. Explicit scenario states and completion rules make intended resistance, unresolved outcomes, safety precedence, and generator fallback testable. Optional generation can improve variation but cannot alter these decisions. Evidence-linked feedback similarly remains an auditable implementation construct, not a validated measure of communication competence. These constraints turn the third research question into a question about specified application behaviour rather than machine empathy or psychological validity.

Finally, evaluation must match each claim. Model experiments assess annotated-label prediction and calibration; authored policy cases assess rule conformance; application tests assess functional, persistence, recovery, accessibility, interface, and performance behaviour; and a limited pilot can describe completion and participant perceptions. Passing one layer cannot substitute for another. In particular, a small, single-arm pilot cannot establish that affect adaptation caused improvement or that learning transfers to real conversations. This layered strategy addresses the fourth research question while preserving the distinction between technical verification and human benefit.

The resulting position is deliberately narrower than a new universal emotion model or an unconstrained role-playing companion. AffectLab investigates whether causal conversational context and text–audio fusion improve a defined speaker-independent task, whether probability quality changes the operational interpretation of those predictions, and whether a complete rehearsal application can apply bounded adaptation while remaining testable under failure. Chapter 3 translates these implications into authority boundaries and requirements; Chapter 4 describes the mechanisms; and Chapter 5 evaluates each evidence layer.

Chapter 3

Proposed System Architecture

3.1 Architecture and Design Principles

AffectLab is a text-first application for rehearsing difficult interpersonal conversations. Its architecture combines deterministic application logic, trained affect-recognition models, optional generative services, and a browser interface, but it is not an end-to-end conversational model. The central design principle is separation of authority: probabilistic components may provide affective evidence or flexible wording, whereas explicit policies retain control of safety, dialogue progression, completion, and feedback evidence. This division makes consequential decisions inspectable and allows the core rehearsal to continue when a model or provider is unavailable.

Figure 3.1 shows the resulting information and authority flow. The interaction layer supports scenario selection, text and optional voice input, conversation display, and reflection. The application layer authenticates the request, validates ownership, coordinates the active session, and persists its state. Within the decision boundary, safety and dialogue policies are authoritative; affect inference supplies bounded evidence to the dialogue policy but cannot replace either authority. Response realisation converts a selected action into text using a constrained generator or its deterministic template. Feedback and reflection operate on recorded policy evidence rather than on unverified generated claims.

The architecture is deliberately text-first. Every conversational turn contains text, either typed by the user or reviewed after optional transcription. Text therefore remains available to the safety and dialogue components even when microphone permission is denied, audio is invalid, trained inference cannot be loaded, or an external provider fails. Raw audio is processed transiently and is not retained as part of conversation history. Optional services can improve modality coverage or linguistic variety, but their loss must not remove the core workflow.

The three standard scenarios—workload negotiation, personal boundary-setting, and expression of a relationship need—share a structural contract. Each defines its objective, stages, difficulty-dependent resistance, observable communication features, response actions, and possible outcomes. A deterministic policy evaluates the current message against that contract and records both the selected action and its evidence. The response-realisation layer receives the action only after this decision; consequently, generated wording cannot redefine success, remove intended resistance, alter the evidence record, or override a safety interruption.

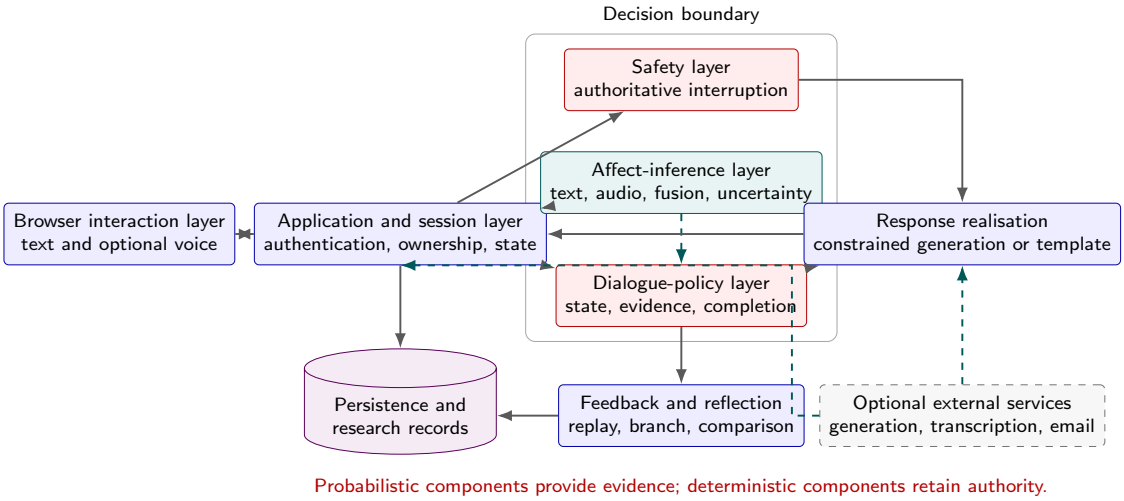


Figure 3.1: Logical architecture of AffectLab and separation between probabilistic evidence and authoritative application control.

Persistence stores the textual turns and enough decision metadata to reconstruct the interaction: state transitions, detected features, selected actions, reason codes, affect-policy decisions, component versions, and generation or fallback provenance. Completed sessions can then support evidence-linked feedback, replay, retry, branching, comparison, personal takeaways, and action cards. Research records form a separate, minimised projection rather than a copy of the participant-facing conversation.

Component boundaries are expressed through narrow contracts rather than shared access to model or provider internals. Inference returns probabilities and provenance; the affect policy returns a permitted adaptation decision; the dialogue policy returns state, evidence, and an action; and response realisation returns wording plus its source. Explicit unavailable and rejected states are part of these contracts. This makes failure behaviour testable, prevents an optional component from silently acquiring authority, and permits models, persistence implementations, or external providers to be replaced without rewriting the scenario policies.

These boundaries align the architecture with the research questions. Contextual text, speech, fusion, and calibration support RQ1 and RQ2; the separation of affect evidence from dialogue and safety authority addresses RQ3; and the integrated, recoverable application provides the object evaluated for RQ4. Model accuracy, policy conformance, application quality, and pilot feedback therefore remain distinct forms of evidence.

3.2 Grouped System Requirements

The original specification contains 29 functional requirements (FR1–FR29) and 14 non-functional requirements (NFR1–NFR14). Table 3.1 consolidates them into nine architectural groups while retaining every identifier used by the implementation and the Chapter 5 traceability analysis. This grouping is sufficient for the design argument; the complete atomic wording is reproducibility material designated for Appendix B during the appendix reorganisation.

Table 3.1: Grouped functional and non-functional requirements.

Group	**Identifiers**	**Required capability or quality**
Identity and data	FR1–FR3, FR26; NFR3–NFR5, NFR13	Authenticated ownership, clear disclosure and preferences, user-controlled optional features, minimised retention, deletion, security controls, and consistent stored state.
Preparation	FR4–FR8, FR19	Scenario preparation and configuration, text-first lifecycle control, and explicit successful, unresolved, manual, maximum-turn, and interrupted outcomes.
Voice and affect	FR9, FR11–FR14; NFR2, NFR14	Reviewed transcription, causal text context, separate audio evidence, declared late fusion, graceful missing-modality handling, and uncertainty-aware bounded adaptation.
Safety	FR10; NFR1	Textual safety checks interrupt ordinary role-play independently of affect classification and before generation.
Dialogue	FR15–FR18; NFR6, NFR8, NFR11–NFR12	Deterministic state transitions and evidence capture, constrained response wording, equivalent fallbacks, auditability, consistency, modularity, and testability.
Feedback	FR20–FR25	Evidence-grounded feedback, controlled phrasing, replay, retry, branching, compatible-attempt comparison, and user-authored reflection.
Research	FR27–FR28; NFR7	Versioned consent and eligibility, withdrawal-aware progress, pseudonymous minimal exports, and reproducible protocol and schema versions.
Support routing	FR29	Transparent lexical routing outside role-play, with tentative rule matches that have no authority over safety, eligibility, scoring, or scenario outcomes.
Operational quality	NFR9–NFR10	Bounded failures and recovery, measurable responsiveness, keyboard and assistive-technology support, visible status, reduced motion, and responsive layouts.

The groups also distinguish core from optional capability. Authentication, text interaction, scenario policy, safety handling, deterministic response templates, evidence-based feedback, and session management form the core. Voice capture, transcription, trained multimodal inference, generative wording, and research participation are optional. A valid implementation may expose an optional feature only when its configuration and failure path preserve the core requirements.

Consolidation does not weaken the atomic acceptance criteria. Each identifier still denotes a separately testable obligation, while a group states the common architectural boundary through which those obligations are implemented. Chapter 5 therefore traces identifiers to evidence rather than treating a group as a single pass/fail claim. This is particularly important for optional features: successful text fallback does not prove trained-inference quality, and successful provider generation does not replace verification of the underlying deterministic action.

3.3 Dialogue, Affect Adaptation, and Safety Boundaries

Each turn follows an authority-ordered pipeline. Authentication, ownership, input, and stale-state checks occur first. The safety layer then examines the text before affect adaptation or response generation. If a supported crisis-related expression

is detected, ordinary role-play stops, a predefined response is returned, and the interruption is recorded. Safety does not use the four-class emotion classifier: a neutral prediction cannot cancel explicit textual evidence, and a generator cannot suppress the interruption.

When no interruption is required, contextual text inference uses the current utterance and a fixed causal window of preceding turns. If valid speech is available, the audio model produces a separate distribution; calibrated modality outputs may then be combined through the declared late-fusion configuration. Availability, confidence, agreement, and component versions remain visible to the policy. Missing audio is absence of evidence, not evidence about the user's affect.

The affect policy maps this information to a small set of reversible actions. Depending on eligibility, user preference, dialogue state, confidence, and modality agreement, it preserves baseline behaviour, offers a pacing choice, or permits a tentative acknowledgement. It cannot determine crisis handling, scenario completion, feedback scores, study eligibility, or research inclusion. Thresholds are operational settings rather than psychological boundaries, and the interface must not present a predicted category as knowledge of an internal state.

In parallel, the scenario policy identifies observable features such as a specific request, maintained boundary, non-blaming statement, acknowledgement, or practical next step. It selects the next action and decides whether the exchange advances, continues under resistance, closes unresolved, reaches a defined outcome, or stops at the turn limit. A generator may phrase the selected action but cannot change it; invalid or unavailable generation invokes the template for the same action. Feedback uses the recorded features and supporting turns, with optional generation limited to restating that authoritative record.

Outside active role-play, deterministic lexical rules may select a bounded support strategy. Labels such as intent, cause, readiness, resistance, or cognitive pattern denote which authored rule fired, not validated psychological constructs. They remain contestable, have no authority over the rehearsal or research workflow, and are evaluated only for rule conformance.

The same authority order governs retries and concurrent requests. A turn is accepted only against the expected session state; a stale update cannot overwrite a newer transition. Retry and branch operations copy declared context but create distinct continuation records, so later feedback can separate inherited turns from new evidence. These constraints make replay reproducible and prevent a delayed provider response from retroactively changing a dialogue decision that has already been persisted.

3.4 Ethics, Privacy, and Design Limitations

AffectLab is intended for voluntary adult rehearsal of ordinary interpersonal conversations. It is not a diagnostic, therapeutic, crisis-monitoring, employment-assessment, educational-assessment, or surveillance system. Completing a scenario means satisfying its declared simulated conditions; it does not establish that the response is appropriate in every real relationship or that the user has improved a general communication skill.

The most important ethical constraint is that annotated expression is not di-

rect access to emotion. IEMOCAP contains ten actors in controlled English-language interactions, and its categories simplify ambiguous affective experience [BBL+08]. Speaker-independent evaluation limits identity leakage but does not establish demographic, cultural, device, or deployment validity. Confidence and text–audio agreement also do not prove correctness, especially under distribution shift [GPSW17, OFR+19]. The architecture therefore confines predictions to low-consequence choices, preserves a non-adaptive path, and allows the user to decline voice or adaptation without penalty.

Privacy constraints follow purpose limitation and data minimisation [Eur16]. Microphone use requires an explicit action; transcripts are reviewable before submission; raw audio is not intentionally retained; conversation storage is time-bounded and account-owned; and research exports exclude direct identifiers, raw conversations, audio, feedback prose, custom scenario wording, and personal takeaways. Pseudonymous records remain potentially linkable while withdrawal is supported and must not be described as anonymous. External transcription or generation also creates a provider boundary that requires disclosure, protected credentials, and a text-only fallback.

Safety remains deliberately limited. Conservative English lexical checks can miss indirect, coded, misspelled, sarcastic, or multilingual expressions and can interrupt quoted or hypothetical language. The application does not know the user's location, provide continuous monitoring, perform clinical triage, or contact emergency services. These limits must accompany the interruption response and prevent the application from being presented as a substitute for professional support.

Fairness and accessibility also shape the design. Text must remain a complete route through the application so that an accent, transcription error, unavailable microphone, or preference not to speak does not remove core functionality. Keyboard operation, visible focus, labelled controls, status announcements, reduced-motion support, and responsive layouts are architectural requirements rather than interface polish. Even so, aggregate model scores and automated interface checks cannot establish equitable performance or accessibility across cultures, communication styles, disabilities, and devices; representative evaluation remains necessary.

These safeguards are design constraints, not evidence of legal compliance. Processing purpose, institutional role, jurisdiction, provider terms, and recipients of affect outputs can change the applicable obligations. In particular, voluntary rehearsal does not authorise employers or educational institutions to use individual emotion inferences for assessment or discipline [Eur24]. Any such change would require a new ethical and legal analysis rather than reuse of the present evaluation.

Deterministic dialogue and feedback improve reproducibility but recognise only anticipated states and language patterns; valid paraphrases, indirect communication, and culturally specific wording may be missed. Generative wording remains probabilistic despite validation. Accessibility checks cannot establish usability for every disabled user, and technical safeguards cannot guarantee confidentiality in an incorrectly configured or compromised deployment. The principal empirical limitations—benchmark validity, authored test coverage, local performance scope, and the non-adaptive pilot—are consolidated with the results in Chapter 5. Accordingly, deployment still requires target-environment security, privacy, accessibility, provider, model, and legal review; the dissertation does not claim compliance certification.

Chapter 4

Implementation

4.1 Technology and Application Overview

AffectLab is implemented as a modular web application whose trained models advise, but do not control, the rehearsal workflow. A React and TypeScript client presents preparation, role-play, feedback, and reflection views. A Python and FastAPI backend owns authentication, dialogue state, safety decisions, feedback evidence, and persistence in MongoDB. Contextual text and speech classifiers can add calibrated affect estimates, while optional provider services can transcribe audio or realise an authorised response in natural language. Each optional component is behind a narrow interface, so the text-based deterministic workflow remains available when model artefacts, credentials, or network services are absent.

This separation is also reflected in deployment. The browser submits typed requests and replaces its local view with the version committed by the backend; it does not calculate authoritative transitions or scores. Backend routes validate transport objects, application services implement policy, and a repository interface supplies either MongoDB persistence or an in-memory substitute for isolated demonstrations. Private datasets, model weights, credentials, and transient audio remain outside version-controlled application code. Dependency sets separate the core web service from the larger local-inference runtime, and containers package the compiled client, API, and database without embedding private artefacts.

Typed boundaries carry the distinctions through the application. A conversation turn can retain the emotion distribution available at that instant, the selected support or character action, the response source, and the pacing decision, while a role-play decision additionally retains its policy version and state snapshots. Absence is represented explicitly rather than as a zero-valued observation. This permits later replay and evaluation to distinguish deterministic, provider-backed, model-assisted, and failed paths without reconstructing them from a transcript.

The implementation is text-first and failure-tolerant. Dialogue progression, evidence collection, crisis interruption, feedback, replay, and session management do not require a graphics processor or external provider. Probabilistic outputs pass through a bounded pacing policy. A language model may phrase a response or feedback item, but it cannot choose an action, declare success, create evidence, or override safety. These authority boundaries organise the mechanisms below.

4.2 Emotion-Recognition Pipeline

The model pipeline converts a private IEMOCAP release into separate text and speech classifiers and then packages compatible full-data artefacts for optional application inference. Preprocessing joins annotations, transcripts, and sentence-level waveforms by utterance identifier, rejects duplicates or missing inputs, and records source and manifest digests. The principal task maps anger, happiness plus excitement, neutrality, and sadness to a fixed four-class order, producing 5,531 matched text–audio utterances. Licensed media and row-level derived data are kept outside the repository. An exploratory six-class mapping is retained only as a secondary research configuration and is not used by the deployed multimodal service.

Figure 4.1 distinguishes evaluation from deployment packaging. Five speaker-independent folds are constructed from complete IEMOCAP sessions: one session is held out for testing, one for validation, and three for training. Because each actor occurs in one session, preprocessing can verify that speakers do not cross split boundaries. These folds provide the evidence reported in Chapter 5. Only after the experimental configuration is frozen are final text and audio models trained on all eligible examples for application use; their training outputs are not treated as a new held-out result.

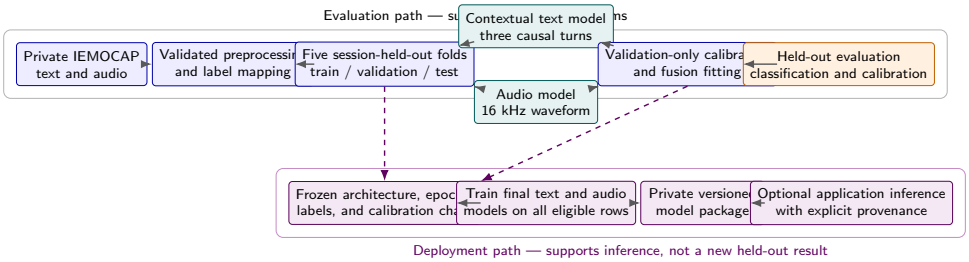


Figure 4.1: Emotion-recognition evaluation and deployment pipeline. Full-data packaging creates application artefacts, not additional held-out evidence.

The contextual text classifier fine-tunes `microsoft/deberta-v3-small`, a compact DeBERTaV3 encoder [HGC21]. Its utterance-only configuration receives the target transcript, whereas the contextual configuration prepends at most three earlier turns from the same dialogue. Each earlier turn is marked as spoken by the same or other speaker relative to the target, and the target receives its own marker. Future turns, emotion annotations, and dialogue outcomes are excluded. The backend reconstructs the same causal format from the stored conversation, avoiding a mismatch between training and inference context.

The acoustic classifier fine-tunes `facebook/wav2vec2-base` on raw waveforms [BZMA20]. Audio is converted to mono, resampled to 16 kHz, limited to 20 seconds, and dynamically padded within a batch. The feature encoder is frozen while the transformer and classification head are trained. Text and audio artefacts share the same label order, and their prediction files must have identical utterance identifiers before fusion is permitted.

Both classifiers are selected by validation macro F1 rather than test performance. Text fine-tuning uses cross-entropy, a small learning rate, warm-up, weight decay, and early stopping; speech fine-tuning adds gradient accumulation and checkpointing to fit the raw-waveform model within available memory. Each fold records

its seed, session assignment, label order, training history, environment, validation calibration, and row-level test distributions. The fusion program verifies these identifiers and reconstructs the saved single-modality metrics before combining probabilities, preventing a result from being produced from mismatched rows or class orders.

Decision-level late fusion keeps the modalities independently trainable and inspectable [BAM19]. For text logits z_t and audio logits z_a, validation-fitted temperatures first produce $p_t = \mathrm{softmax}(z_t/T_t)$ and $p_a = \mathrm{softmax}(z_a/T_a)$. The distributions are then combined as

$$p_m = wp_t + (1 - w)p_a, \tag{4.1}$$

with a predeclared $w = 0.5$ for the principal research comparison. A separate deployment path chooses w and a final temperature from validation-role predictions only. Temperature scaling changes probability concentration without changing the largest logit [GPSW17]; classification and calibration are consequently evaluated as separate properties. The packaged global parameters are operational settings rather than independent performance evidence because their pooled fitting procedure reuses utterance labels across rotating fold roles.

At inference, the backend returns complete calibrated distributions, selected labels, confidence, modality agreement, and separate queue and execution times. Confidence below 0.55 is presented as low, from 0.55 to below 0.75 as moderate, and at least 0.75 as high. These are operational display and policy thresholds, not psychological boundaries. Missing audio, disagreement, invalid distributions, or unavailable models remain explicit; they are never converted into a stronger estimate. Calibration can also deteriorate under shifts in speaker, language, device, and interaction style [OFR+19], so the application treats the distributions as uncertain evidence rather than descriptions of internal state.

The private inference package contains the two models, their tokenizer or feature extractor, the fixed label order, input limits, and the complete calibration chain. A provenance manifest binds these files to source and configuration hashes without publishing licensed weights. Models load lazily and run behind an asynchronous lock to bound memory contention. Multimodal inference is disabled by default and failure returns the exchange to the deterministic text path.

4.3 Backend and Request Processing

The FastAPI backend is the authoritative coordination layer. API routes and Pydantic schemas constrain inputs and outputs, application services coordinate authentication, conversation, model, and provider functions, and repositories encapsulate mutable state. This layering keeps transport logic separate from dialogue policy and permits deterministic or trained implementations to satisfy the same interfaces. Session documents retain ordered turns, role-play snapshots, evidence, policy decisions, feedback, model metadata, version, and expiry time; response schemas expose only what each interface requires.

Authentication and routine account operations use the same service boundary but are not part of the research contribution. Passwords and verification or recovery tokens are stored only as adaptive hashes or digests. Short-lived access tokens

authorise API requests, while rotated refresh tokens use HTTP-only cookies. More importantly for the rehearsal data, every session lookup includes both the session and authenticated owner identifiers. Each write also supplies an expected monotonically increasing version and a request identifier; stale or duplicate submissions return a conflict rather than silently overwriting a committed turn. MongoDB provides durable storage and expiry indexes, while the in-memory repository implements the same contract for controlled local use.

Request schemas enforce message, title, reflection, and encoded-audio limits before service execution. Enhanced exchanges must pair their expected version with a unique request identifier, and received probability vectors are accepted only when all values are finite, bounded, normalised, and consistent with the declared leading label. The backend derives agreement and confidence bands again instead of trusting client-authored summaries. These checks keep a malformed enhancement from contaminating the authoritative session even when ordinary text processing can continue.

Figure 4.2 shows the processing order for a conversation exchange. After validation, authentication, ownership, duplication, and version checks, a local crisis-language detector runs before affect inference or response generation. A safety match interrupts the rehearsal through fixed wording and skips optional processing. Otherwise, eligible requests may obtain a multimodal estimate; failure is recorded as unavailability and does not abort the message. The service then appends the user turn, advances the deterministic controller, records evidence, applies any permitted pacing action, realises the authorised reply, and persists the complete transition atomically. The response contains the committed version and server-owned state so the browser does not need to infer what succeeded.

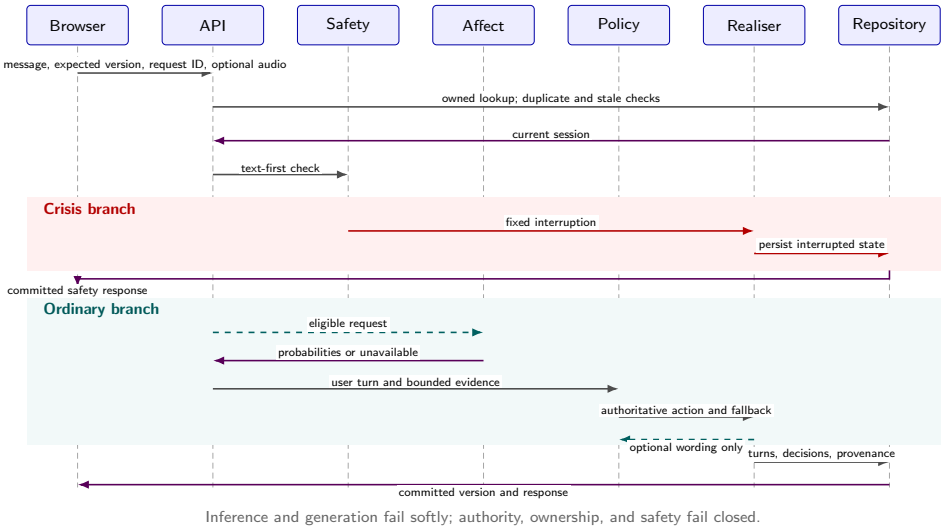


Figure 4.2: Version-bound conversation request with safety precedence and failure-tolerant optional processing.

Provider-backed response realisation follows the same authority rule. It receives the selected support strategy or role-play action and must return a typed, short response consistent with that action. Empty, refused, malformed, unsafe, or contradictory output is discarded in favour of deterministic wording. A small rule-based analyser can expose tentative lexical intent, possible patterns, and authored

support priorities outside active role-play, but these values are rule matches rather than learned probabilities or validated psychological measurements. During role-play, neither these heuristics nor provider output may advance the scenario, create evidence, or determine success.

4.4 Role-Play and Affect-Adaptation Engine

A scenario is represented by its participant objective, character, opening line, supported observable skills, difficulty behaviour, completion conditions, and maximum turn count. Standard scenarios cover workload prioritisation, maintaining a boundary, expressing a relationship need, workplace feedback, deadline negotiation, and shared responsibilities; the first three form the fixed pilot sequence. Custom scenarios reuse only supported features and present any prepared objection as an editable rehearsal prompt, not as a prediction of another person.

The enhanced workload, boundary, and relationship scenarios are implemented as versioned state graphs. Workload practice moves from describing the problem through acknowledging constraints to a workable plan. Boundary practice requires a clear refusal to survive authored pressure before respectful closure. Relationship practice requires a specific need, acknowledgement of perspective, and a practical request. A transition occurs only when the evidence required at the current stage is present; contradictory or unsupported input leaves the controller in place. Attempts can also end at the turn limit, by manual finish, or through safety interruption, and each decision stores its before-and-after snapshot so a restored session does not reconstruct state from prose.

Difficulty controls authored resistance and cooperation rather than changing what a detected feature means. Eligible additional practice can also select a co-operative, rushed, or sceptical character profile, each backed by a versioned policy; profiles are excluded from required pilot tasks. Every processed rehearsal turn records the chosen action, reason codes, linked user and assistant turns, supporting evidence identifiers, generation source, and scenario, policy, and scoring versions. Completion reasons remain distinct, allowing the same persisted representation to support recovery, audit, and fair comparison.

Evidence extraction records bounded language features such as a concrete request, specific detail, I-statement, non-blaming wording, maintained boundary, acknowledgement, or feasible proposal. Each item links to its source turn, and features may accumulate only where the scenario policy permits. The rules account for selected negations and contradictions but cannot cover every indirect, sarcastic, multilingual, or complex expression. A match therefore means that an authored rule observed a feature in this rehearsal; it is not proof of intent, competence, or probable real-world success.

The affect-pacing policy in Figure 4.3 is applied only after the controller has selected an action. Safety, dialogue state, and explicit participant preference take precedence. Automatic adaptation is disabled by default and is limited to active, enhanced, non-required rehearsals. Required pilot tasks, retries associated with those tasks, paused or completed attempts, and unsupported scenarios remain on the baseline path. The participant can independently request unchanged, gentler, or more challenging pacing, so the control remains usable without voice.

When automatic adaptation is eligible, missing audio, inference failure, or an unsupported category produces no addition. Low confidence in any component or disagreement between fused, text, and audio leaders produces a neutral pacing choice. Only when all three labels agree and meet the low-confidence boundary can anger or sadness add the fixed acknowledgement "I hear you."; happiness or neutrality adds nothing. The policy never translates a label into anxiety, resistance, readiness, or a clinical state, and it cannot change stage, evidence, success conditions, or feedback scores. The stored decision includes the selected action, reason, preference, model and policy versions, timing, and distributions, allowing the exact behaviour to be explained after reload.

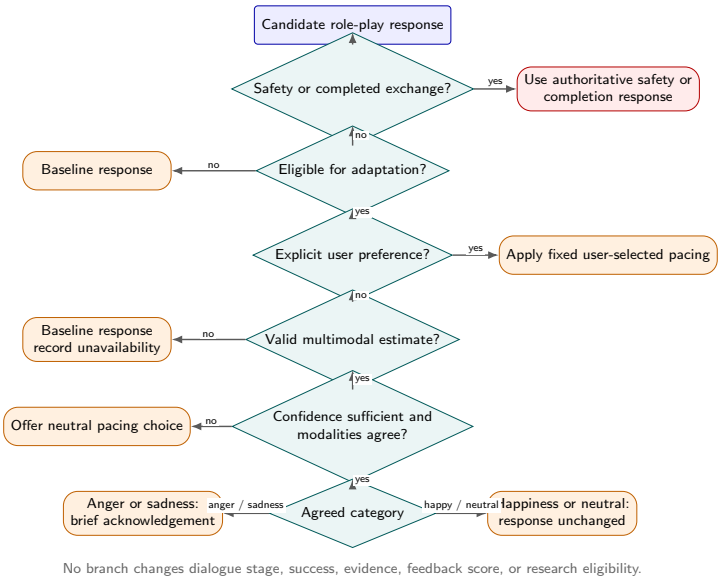


Figure 4.3: Bounded affect-aware pacing. Every terminal action preserves the authoritative dialogue state and evidence.

Response wording begins with the action and fallback supplied by the deterministic controller. If generation is enabled, the provider must repeat the required action, stay in character, avoid coaching language, and preserve any concrete negotiated constraint. Parsing failure, action mismatch, disallowed content, or loss of a material fact restores the fallback. Branches also use deterministic wording so their differences arise from the participant's alternative response rather than uncontrolled regeneration. Safety uses an independent local detector and fixed interruption path before either mechanism; it is an engineering safeguard, not a validated clinical risk assessment.

4.5 Feedback and Frontend Workflow

Feedback is calculated from the evidence and decisions stored during the rehearsal rather than from a new model judgement. Scenario-specific metrics refer to supported features and their source turns; deterministic templates convert them into observed strengths and next-attempt suggestions. The completion reason remains separate, so reaching a turn limit is not relabelled as success and a safety interruption is not scored as a skill failure. Optional provider phrasing may reword

an already calculated item but cannot alter measurements or invent observations. Comparisons with earlier attempts are made only under the same scoring version and are described as differences in observed rehearsal language.

Replay is a read-only projection of the recorded exchanges, state transitions, generation source, linked evidence, and saved pacing decision. Rewind restores a preceding stored snapshot only when the latest user–assistant pair can be removed safely. Branching instead creates a copy-on-write child from a validated pre-turn snapshot; copied context remains distinguishable from the new continuation and is not counted again in feedback. The original attempt is unchanged, and comparison presents both continuations without ranking either as globally better. A personal takeaway and editable action card remain user-authored reflection rather than evaluation evidence.

Measurement boundaries prevent later reflection from entering rehearsal feedback. Missing legacy decisions or model output are displayed as gaps rather than regenerated. A branch validates source ownership, version, policy compatibility, and the selected pre-turn snapshot, then stores its lineage and copied-turn boundary. Action-card suggestions reuse only eligible participant turns inside the measurement window; safety-interrupted content and copied branch context are excluded, and every field remains editable before saving.

The React interface follows the same lifecycle: preparation, active rehearsal, feedback, replay or branching, and an action card. The representative role-play view in Figure 4.4 combines the transcript and composer with scenario progress and optional voice controls. The browser includes the displayed session version and a fresh request identifier, clears a draft only after confirmation, and offers reload after a conflict while preserving unsent text. Semantic controls, visible focus, live status regions, reduced-motion rules, and responsive layouts support keyboard, desktop, and mobile use.

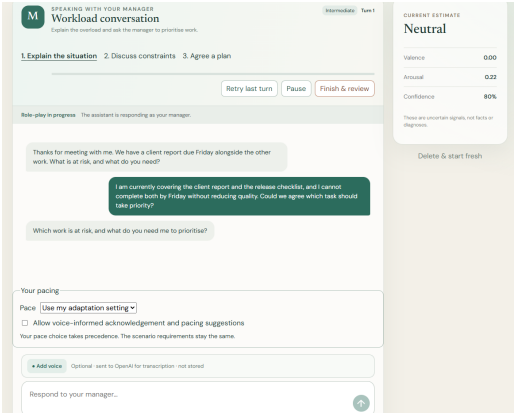


Figure 4.4: Representative role-play workspace with transcript, scenario state, message composer, and optional voice interaction.

Voice input is opt-in for each exchange. The browser requests microphone access only after an explicit action, records at most 20 seconds, converts the signal to mono 16 kHz PCM WAV, and keeps it in component memory. A transcription result enters the composer for review rather than becoming a turn automatically. The reviewed text and optional audio are then sent together so prediction, pacing, and stored wording refer to one exchange. The backend validates the audio and processes it in

memory; raw recordings are not placed in sessions or research exports.

The interface distinguishes the trained multimodal distribution from a separate deterministic lexical display. When multimodal output exists, it shows fused confidence, modality leaders, agreement, the full distribution, timings, and an audio-not-stored notice. Disagreement and unavailable inference remain visible. Reloading uses the persisted decision rather than rerunning the model, preventing a later estimate from being presented as the reason for an earlier response.

Optional research support is limited to a versioned pilot workflow: explicit consent and eligibility, three ordered required tasks, immediate questionnaires, pseudonymous progress records, and a text-free export. Required tasks deliberately bypass multimodal inference and affect-aware pacing. The workflow therefore supports collection for the baseline deterministic rehearsal only; its implementation does not make the pilot evidence about adaptation. This distinction is carried into the evaluation in Chapter 5.

4.6 Security, Persistence, and Deployment

Security controls concentrate on ownership, minimisation, and explicit failure boundaries. MongoDB queries bind session identifiers to their owner; optimistic versions prevent lost updates; production configuration rejects weak signing secrets, wildcard origins or hosts, disabled HTTPS enforcement, and disabled rate limiting. Conversation text persists because continuation, replay, and evidence links require it, but raw audio is transient. Research projections exclude identity, transcripts, free-text feedback, custom wording, and audio and use a retention schedule separate from ordinary sessions. Account deletion cascades through owned operational records, while study withdrawal removes live research records without pretending that irreversibly de-identified frozen exports can still be linked.

Operational and research records have different purposes and expiry fields. The richer account-owned session supports continuation and user-controlled deletion; the pseudonymous study projection carries protocol versions, task classification, questionnaire values, numeric feedback, and event counts without copying conversation content. Freezing creates a checksum-bound de-identified export and closes further protocol mutations. This lifecycle keeps pilot administration from silently expanding the information retained for ordinary practice.

The integrated deployment uses a compiled frontend, FastAPI service, and MongoDB with health checks. Readiness verifies configuration, database access, and required indexes; missing optional transcription or model capability is reported separately and does not make the text-first application unavailable. Expected authentication, validation, rate-limit, provider, model, and concurrency failures become bounded client responses. Authority failures—missing ownership, stale state, invalid consent, or a frozen protocol—fail closed, whereas enhancement failures return to deterministic operation.

Verification evidence is reported once in Chapter 5. Unit, service, browser, and full-stack checks exercise the compiled client and real database while external providers and trained inference remain disabled; performance therefore describes only that controlled local path. Reproduction additionally requires the source revision, locked dependencies, versioned configurations, and private artefact hashes.

Chapter 5

Evaluation and Results

5.1 Evaluation Methodology

The evaluation separates three kinds of evidence because they support different conclusions. First, speaker-independent IEMOCAP experiments address whether dialogue context improves text emotion classification (RQ1) and whether acoustic information improves a contextual text model (RQ2). Second, authored policy cases and application tests address whether uncertain predictions are translated into the specified bounded actions (RQ3) and whether the integrated system satisfies its declared requirements under reproducible developer-led verification (RQ4). Third, a small pilot provides supplementary feedback about the baseline rehearsal workflow. It does not evaluate emotion-aware adaptation because the required pilot tasks bypassed multimodal inference and affect pacing.

This hierarchy prevents benchmark accuracy, deterministic policy conformance, software verification, and user response from being collapsed into a single claim. IEMOCAP measures classification and probability quality on a controlled corpus; application tests establish behaviour only for their exercised paths and environments; and the pilot describes immediate experience only for the workflow participants encountered. Deployment readiness consequently remains conditional on target-environment validation.

5.1.1 Evidence Scope and Decision Rules

Each research question has a distinct evaluation unit. For RQ1, the unit is an IEMOCAP utterance paired with its causal dialogue history, and the admissible conclusion is a difference in benchmark text classification on identical held-out sessions. For RQ2, the unit is a matched text–audio utterance and the conclusion concerns complementary predictive information and calibration on the same benchmark. For RQ3, the unit is a versioned application decision within a controlled workflow; the tests may establish that the deterministic policy followed its specification, but not that a label was psychologically correct or that the action helped a person. For RQ4, the unit is the integrated application and its principal workflows, so several complementary checks are required across service, persistence, browser, accessibility, recovery, and timing boundaries.

The comparisons were chosen before interpreting the final pooled results. Utterance-only text is the RQ1 reference, and three-turn contextual text is the

RQ2 reference. Predeclared equal fusion is the primary classification treatment; validation-fitted fusion answers the different question of whether operational probability quality can be recovered. The pilot is not promoted to a fifth evaluation level because its required tasks did not enter the multimodal path. It instead provides descriptive evidence about workflow completion and participant ratings for the deterministic baseline.

Evidence is reported at the narrowest defensible level. A passed test denotes observed conformance, not proof of correctness for all inputs. A bootstrap interval concerns the sampled dialogue clusters, not unseen populations. A zero-violation accessibility scan concerns automated rules and rendered states, not conformance certification. Similarly, a local latency distribution is a regression measurement rather than a production service-level result. These decision rules are applied consistently in the results and threats-to-validity sections.

5.1.2 Benchmark Task and Speaker-Independent Protocol

The primary task uses the IEMOCAP categories anger, happiness, neutral, and sadness, with excitement merged into happiness and other source labels excluded [BBL+08]. It contains 5,531 matched text–audio utterances from 151 dialogues across five dyadic sessions. Text, audio, and fusion experiments use identical utterance identifiers and label order.

Five folds provide speaker-independent coverage. One complete session is held out for testing and another for validation in each fold; the remaining three sessions are used for training. Every session appears exactly once as test data, and both speakers in a dyad stay together. This design prevents a tested speaker's utterances from entering that fold's training set. Validation predictions select temperatures and, for the calibration-oriented fusion variant, fusion parameters. Only the held-out session supplies evaluation predictions.

The utterance-only and contextual text conditions use the same DeBERTa-v3-small architecture, folds, labels, seed, and optimisation settings. Their intended difference is the input: the contextual condition prepends the preceding three causal dialogue turns with speaker-relative markers. The paired audio condition uses wav2vec 2.0 on the 16 kHz waveform. The equal-weight fusion rule was fixed before inspecting combined test outcomes and is therefore the principal classification comparison. Fold-specific validation-fitted fusion is analysed separately as a probability-quality configuration.

Final application packages were trained on all eligible utterances after the cross-validation design and epoch counts were frozen. They have no independent test partition. Their training and smoke checks are integrity evidence rather than performance estimates; all model-performance claims below come from the five session-held-out experiments.

The separation of development roles is especially important for calibration. A temperature or fusion weight can appear effective if it is fitted and assessed on the same labels, even when the underlying classifier is unchanged. The fold-specific chain avoids that circularity by estimating every parameter from the validation session before applying it to the unseen test session. By contrast, the later global chain necessarily uses all available validation-role labels to prepare one deployable package and is reported only as a descriptive packaging check. This distinction

allows the dissertation to discuss an operational configuration without presenting its label-reused diagnostic values as a second independent experiment. Engineering fixtures are separated for the same reason: their controlled probability vectors test policy branches, but they neither train the classifiers nor enter the IEMOCAP performance totals.

5.1.3 Measures and Statistical Comparison

Accuracy, macro F1, and weighted F1 are reported for classification. Macro F1 is primary because it weights the four classes equally despite unequal support. Probability quality is evaluated separately through expected calibration error (ECE), negative log-likelihood (NLL), and multiclass Brier score. ECE summarises the gap between confidence and accuracy within bins, but it is bin-dependent and incomplete; NLL and Brier score therefore provide complementary proper scoring rules [GPSW17, KPNK+19]. Temperature scaling is fitted to validation logits only:

$$\hat{\mathbf{p}} = \mathrm{softmax}\left(\frac{\mathbf{z}}{T}\right), \qquad T > 0. \tag{5.1}$$

A scalar temperature changes probability concentration without changing the predicted class.

Because utterances from one conversation are dependent, comparisons use a paired percentile bootstrap over the 151 complete dialogues. In each of 10,000 replicates, dialogues are sampled with replacement and the same sampled clusters are applied to both systems before recomputing each aggregate metric. The 2.5th and 97.5th percentiles form the 95% interval; the procedure uses seed 20260808. These intervals describe resampling uncertainty among the observed dialogues, not demographic, cultural, linguistic, device, or deployment generalisation.

Pooling the five non-overlapping test sessions produces one prediction for every eligible utterance and supplies the headline scores. Fold-level values are retained separately to show whether an aggregate result is driven by one session, but their unweighted mean is not interchangeable with a pooled nonlinear metric such as macro F1. No speaker-subgroup inference is attempted: IEMOCAP's ten actors provide too little support for such analysis. Per-class metrics are likewise descriptive unless their paired dialogue-bootstrap interval is reported.

5.2 Contextual Text Results

Table 5.1 gives the primary RQ1 comparison. Three-turn causal context improved accuracy from 0.6624 to 0.7304, macro F1 from 0.6651 to 0.7332, and weighted F1 from 0.6640 to 0.7300. The macro-F1 gain was 0.0681, with a dialogue-bootstrap 95% interval of 0.0504–0.0857; all three aggregate intervals excluded zero.

Table 5.1: Four-class contextual-text results on five held-out IEMOCAP sessions.

Metric	Utterance only	Three-turn context	Difference	95% interval
Accuracy	0.6624	0.7304	+0.0680	[0.0496, 0.0864]
Macro F1	0.6651	0.7332	+0.0681	[0.0504, 0.0857]
Weighted F1	0.6640	0.7300	+0.0661	[0.0477, 0.0842]

The class effects were uneven. Happiness, neutral, and sadness F1 increased by 0.0882, 0.0338, and 0.1307, respectively, with dialogue-bootstrap intervals above zero. Anger F1 increased by 0.0198, but its interval of $[-0.0073, 0.0449]$ included zero. The result therefore supports an aggregate contextual benefit, not an anger-specific improvement. The calibrated contextual model obtained ECE 0.0311 on the benchmark; this does not validate AffectLab's operational presentation thresholds on user-authored conversations.

The pattern is consistent with context resolving labels whose interpretation depends on an earlier exchange, while also showing that extra turns are not uniformly beneficial. In one held-out happiness example, an utterance-only neutral prediction changed to the correct class after context was added. Another held-out neutral example changed from a correct local decision to anger, illustrating contextual carry-over. A third neutral example assigned equal posterior mass of 0.4407 to happiness and neutral before the stored argmax selected happiness. These selected cases do not quantify failure frequency, but they show why a changed top label should not automatically be read as a decisive contextual interpretation.

The comparison also isolates only the combined effect of the contextual input representation. It does not determine whether the improvement came from additional lexical content, the identity-relative speaker tokens, sequential emotional consistency, or interaction structure. Component ablations would be required to attribute the gain among those mechanisms. Because both conditions share the same folds and training configuration, however, the aggregate paired difference remains a direct answer to the narrower question asked by RQ1.

An exploratory six-class task also improved from 0.5490 to 0.6222 macro F1, but only 40 of its 7,420 examples were mapped anxiety cases. The result is retained in the prepared Appendix C source with the per-class and fold tables. It is secondary evidence and does not support a deployed anxiety-recognition claim.

Within this protocol, RQ1 is therefore answered positively: causal dialogue history improved aggregate text classification while architecture, data splits, and training settings were held constant. The experiment does not reveal whether the gain comes from additional lexical material, speaker markers, or discourse history, and it cannot show that a predicted category represents a user's internal emotional state.

5.3 Speech, Fusion, and Calibration Results

5.3.1 Audio and Predeclared Fusion

The wav2vec 2.0 audio baseline obtained accuracy 0.6279, macro F1 0.6349, weighted F1 0.6229, and calibrated ECE 0.0468. It was weaker than contextual text overall but stronger for anger recall, suggesting that the branches contained complementary predictive information. This does not mean that the acoustic representation isolated prosody: it may also encode phonetic, lexical, speaker, and recording characteristics.

The principal fusion configuration averages the validation-calibrated text and audio posterior distributions with fixed weights of 0.5. Table 5.2 shows improvements over contextual text on every aggregate classification measure. Macro F1 increased by 0.0551, with a paired 95% interval of 0.0407–0.0713; all 10,000 dialogue-

bootstrap replicates favoured fusion for accuracy, macro F1, and weighted F1. Because the rule was predeclared rather than selected from fused test performance, it provides the headline RQ2 classification evidence.

Table 5.2: Predeclared equal-weight fusion compared with contextual text.

Metric	Context text	Equal fusion	Difference	95% interval
Accuracy	0.7304	0.7832	+0.0528	[0.0381, 0.0683]
Macro F1	0.7332	0.7883	+0.0551	[0.0407, 0.0713]
Weighted F1	0.7300	0.7815	+0.0514	[0.0366, 0.0671]

The pooled class results reinforce that interpretation. Equal fusion reached F1 values of 0.8277 for anger, 0.8163 for happiness, 0.7014 for neutral, and 0.8080 for sadness. Neutral remained the hardest class. Inspection of matched errors showed why fusion is not uniformly corrective: a strong audio error could overturn a correct text decision, and agreement between two wrong branches could reinforce a shared error. These cases are explanatory examples rather than prevalence estimates; complete class and session results are preserved in the prepared Appendix C source.

Matched predictions illustrate both directions. For one anger utterance, contextual text selected neutral while audio selected anger, and fusion recovered the annotated class. For a neutral utterance, text was correct but audio favoured anger strongly enough that fusion narrowly selected anger. In another neutral case, both modalities predicted anger and averaging reinforced the shared error. Late fusion can therefore exploit complementary mistakes but cannot correct correlated ones, and a fixed average can allow an overconfident branch to dominate a better counterpart.

Session-level macro F1 preserved the aggregate ordering. Context exceeded utterance-only text in all five held-out sessions, and equal fusion was the strongest classifier in every session. Audio varied most, with fold scores from 0.5870 to 0.6974 and a descriptive standard deviation of 0.0386; contextual text and equal fusion each had a standard deviation of about 0.0227. These five dyads reveal sensitivity within the corpus but are too few to estimate population speaker variability.

5.3.2 Classification–Calibration Trade-off

Equal averaging achieved the strongest classification but ECE rose to 0.1619. Separately calibrated modalities therefore did not remain calibrated after their probabilities were averaged. This matters operationally because AffectLab consumes confidence when selecting whether to preserve baseline pacing, offer a choice, or permit a bounded acknowledgement.

Fold-specific validation fitting selected text weights between 0.54 and 0.63 and fitted a final temperature. As Table 5.3 shows, this reduced ECE to 0.0283 while retaining accuracy 0.7717 and macro F1 0.7758. Its NLL and Brier score were also the lowest among the independently evaluated configurations. The approximately one-point macro-F1 reduction relative to equal fusion is consequently an explicit trade-off, not an inconsistency between results.

Figure 5.1 places the main classification and calibration results on common scales. The proper scores qualify an ECE-only interpretation: equal fusion was under-confident, yet its NLL and Brier score improved over both unimodal branches.

Table 5.3: Classification and probability quality for independently evaluated fold configurations.

Configuration	Accuracy	Macro F1	Weighted F1	ECE	NLL	Brier
Contextual text	0.7304	0.7332	0.7300	0.0311	0.7061	0.3845
Audio only	0.6279	0.6349	0.6229	0.0468	0.9441	0.5140
Equal-weight fusion	0.7832	0.7883	0.7815	0.1619	0.6843	0.3619
Validation-fitted fusion	0.7717	0.7758	0.7707	0.0283	0.6050	0.3273

Validation-fitted fusion then improved all three probability-quality measures. Classification quality and probability quality must therefore remain separate claims.

The apparently counter-intuitive equal-fusion result follows from the metrics measuring different properties. ECE compares average confidence with empirical accuracy inside bins, whereas NLL and Brier score use the complete predicted distribution. Equal fusion often placed useful probability mass on the correct class while remaining systematically under-confident in its populated mid-confidence bins. Validation-derived weighting and temperature scaling brought maximum confidence closer to observed accuracy without preserving every classification decision. For an application whose policy consumes confidence, the better-calibrated configuration is therefore preferable even though the fixed average has the highest F1.

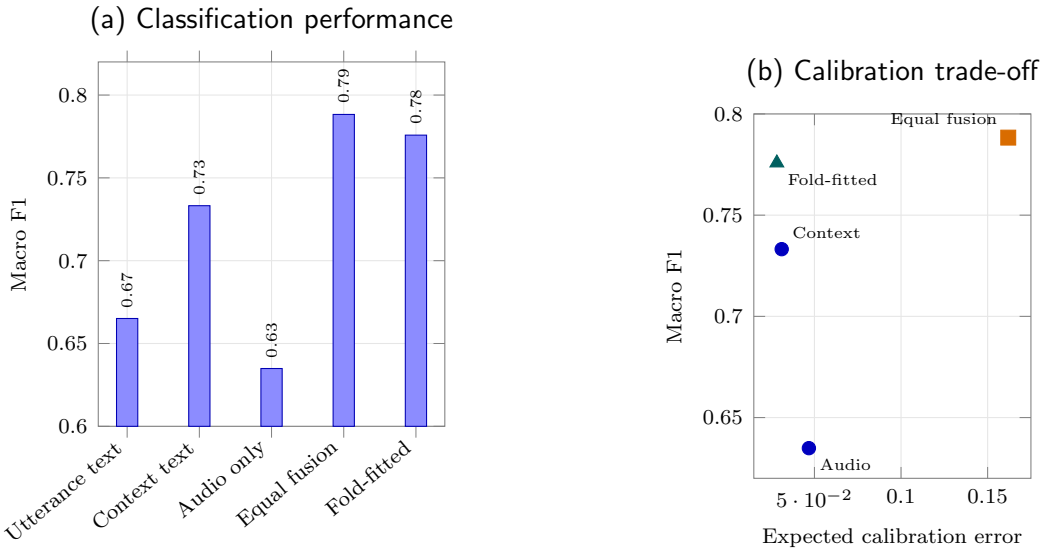


Figure 5.1: Four-class held-out results. Panel (a) compares macro F1; Panel (b) shows the calibration cost of equal fusion and the effect of validation-derived fitting.

The application requires a single chain for models trained on all available data. A global fit selected text/audio weights of 0.59/0.41 and a final temperature of 0.5989. Its descriptive pooled values were macro F1 0.7776 and ECE 0.0242, but labels used in calibration fitting recur in the reporting pool through predictions from different fold models. These are therefore label-reused packaging diagnostics, not additional held-out evidence. The equal-weight result remains the classification headline, while fold-specific validation fitting supplies the independent calibration evidence.

The closest published protocol-compatible result is Li et al.'s four-class, five-session IEMOCAP study, which likewise reports text outperforming audio and multimodal classification outperforming both [LWC+22]. AffectLab's equal fusion

reached 78.32% accuracy compared with their 80.36% multimodal value. This is descriptive placement rather than a head-to-head ranking: their scores average folds, their validation allocation and architectures differ, and their text model uses future as well as preceding context. Six-class and random utterance-split studies are not treated as comparable.

RQ2 is thus answered positively within IEMOCAP: acoustic information improved classification over contextual text on matched held-out examples. The operational answer must include the calibration trade-off, however. Neither benchmark calibration nor the global packaging chain guarantees reliable confidence after domain shift [OFR+19].

5.4 Application Verification

Application evidence addresses conformance of the bounded integration policy and the complete implementation, not psychological or clinical validity. Table 5.4 consolidates the principal results. Detailed requirement traceability, authored cases, test commands, and performance rows are preserved in the prepared Appendix D source.

Table 5.4: Aggregate application-verification evidence.

Evidence	**Outcome**	**Scope boundary**
Authored policy cases	5/5 dialogue trajectories and 7/7 affect-pacing cases matched declared outcomes	Rule conformance on controlled inputs, not comparative helpfulness or human benefit
Diagnostic extraction fixture	5 true positives, 4 true negatives, 1 false positive, 2 false negatives	Twelve designed examples; not an accuracy estimate
Backend and static checks	245/258 backend cases passed; 13 declared real-MongoDB skips; Ruff passed	Default suite used controlled inputs and a memory repository
Frontend and browser	Production build and lint passed; 76/76 desktop/mobile cases passed	Bundled Chromium and two declared viewports
Automated accessibility	20 scans across 10 states returned zero detected axe violations	Automated rules only; not WCAG certification or assistive-technology testing
Integrated stack smoke	2/2 compiled-stack journeys passed, including two backend restarts	Live providers and trained inference disabled
Local performance	5/5 budgets passed; conversation turn median 4.659 ms, P95 5.836 ms	In-process, single client; excludes network, MongoDB, models, providers, and concurrency

5.4.1 Bounded Policy Behaviour

Five authored enhanced-dialogue trajectories covered workload, boundary, and relationship scenarios. All followed their expected controller stages, including non-confirmation and unresolved endings. The cases show that completion requires scenario-specific evidence rather than a generic success phrase. They do not establish that the resulting dialogue is more realistic, persuasive, or educational.

Twelve human-authored judgements tested the lexical evidence extractor. The false positive arose when reported speech containing "I need" was attributed to the current speaker; an indirect refusal and the synonym "shift" produced the two false negatives. Keeping these errors visible demonstrates the limitations of deterministic extraction and motivates replaying the source utterance alongside derived feedback. The counts are diagnostics, not a held-out accuracy rate.

Seven affect cases covered confident agreement, modality disagreement, low confidence, missing audio, unavailable inference, disabled adaptation, and explicit preference. All produced the declared action: confident agreement permitted a brief bounded acknowledgement; disagreement and low confidence offered a pacing choice; failures and disabled adaptation preserved the baseline; and an explicit user preference took precedence. Separate tests verify that safety, completion, study eligibility, and required-task rules override adaptation. The classifier supplies evidence, while deterministic dialogue and safety policies retain authority.

Implementation-authored cognitive and support-routing cases similarly verify only stable rule execution and precedence. They do not validate constructs such as intent, cause, readiness, resistance, or cognitive distortion, nor do they show that a selected strategy is helpful or harmless. This distinction bounds the RQ3 conclusion: AffectLab demonstrably maps uncertain or unavailable evidence into auditable actions and fallbacks under tested conditions, but it does not establish the truth of an inferred state or the benefit of adapting to it.

Persistence and replay make the policy observable rather than merely repeatable in memory. Each decision is bound to an exchange version and stores the modality evidence, confidence state, selected action, and fallback reason needed to reconstruct the outcome. Optimistic concurrency rejects stale updates, preventing a delayed inference result from silently attaching to a newer turn. Replay presents the original utterance beside derived evidence so that a user can inspect the basis of feedback, while branching creates a new history rather than rewriting the original. These mechanisms support explainability as traceability of system action; they do not make the underlying inference inherently correct.

5.4.2 Integrated Quality and Local Responsiveness

The final default backend run collected 258 cases: 245 passed and 13 explicitly skipped because a real-MongoDB URI was absent. The skips belong to one declared environment-dependent integration module and are not counted as passes. Ruff, the production frontend build, and frontend linting passed. A focused browser run expanded 38 logical cases across desktop Chromium and a Pixel 7 profile, producing 76 passes. Twenty axe-core scans covered ten interface states in both profiles and detected no violations under the selected automated WCAG 2.0/2.1 A/AA rule tags. This is evidence about the exercised DOM states, not accessibility certification.

The browser cases covered preparation, active practice, restored reflection, voice-text review, uncertain pacing, stored provenance, replay, branch comparison, and action-card views. They also asserted keyboard interaction, focus behaviour, status announcements, preservation of unsent input after failure, and recovery controls. Testing both projects checks responsive layout at the declared viewports, but it does not sample the wider device, browser, zoom, or assistive-technology space.

Two separately recorded smoke journeys used the compiled frontend, an actual FastAPI process, and disposable MongoDB 8.0.1. They covered account access, consent and study flow, required-task persistence, researcher export, enhanced dialogue, failure recovery, replay, branching, comparison, and action-card persistence. State remained correct across two backend restarts. Trained inference, transcription, and generation providers were disabled, so the journeys establish integration and recovery for the deterministic configuration rather than live-provider behaviour.

The relationship between the default and smoke evidence is complementary. The fast default suite gives broad deterministic coverage but substitutes the memory repository; the two smoke journeys provide narrower end-to-end coverage with real process and database boundaries. The 13 skipped MongoDB tests make this division explicit instead of treating unavailable infrastructure as success. Neither layer independently supports a production claim, but together they show that the application contract survives both component-level variation and the recorded database-backed journeys.

The local regression benchmark exercised FastAPI's in-process HTTP boundary with authentication, validation, services, serialisation, and the memory repository. All five predeclared budgets passed. The deterministic conversation turn had median 4.659 ms and nearest-rank P95 5.836 ms over 100 sequential samples. These numbers characterise locally controlled application overhead only; they exclude network transit, MongoDB, trained inference, provider latency, and concurrent clients.

Together, the authored, automated, browser, persistence, recovery, and timing evidence answers RQ4 for the tested offline deterministic core: the implemented paths conform to their declared functional and non-functional requirements and are suitable for controlled deployment after environment-specific validation. Production claims require staged tests with intended models and providers, real network and database infrastructure, expected concurrency, backup recovery, observability, and independent security assessment.

5.5 Supplementary Pilot Results

The single-arm pilot examined only the baseline text rehearsal with deterministic dialogue and feedback. Every required task was ineligible for multimodal inference and affect-aware pacing. The results below therefore cannot establish whether emotion-aware adaptation is acceptable, accurate, useful, beneficial, or harmful.

The analytic export contained 15 participants. Twelve completed all three required tasks and questionnaires, giving 80.0% protocol completion with a two-sided 95% Wilson interval of 54.8%–93.0%. There were 42 first required attempts, of which 39 reached task completion, and 40 available realism/usefulness ratings. Table 5.5 retains the outcomes needed to interpret this supplementary evidence.

The 38 paired confidence ratings came from 14 participants, and their interval

Table 5.5: Supplementary pilot results for the non-adaptive required workflow. Ratings use seven-point scales.

Measure	n	**Mean**	**Median**	**Additional result**
Scenario realism	40	4.85	5	–
Feedback usefulness	40	5.33	5	–
Paired confidence change	38	+0.97	+1	95% interval +0.67 to +1.26

was computed by participant-cluster bootstrap. This immediate self-report change is not independently measured communication improvement. Two required attempts lacked a post-questionnaire, denominators vary with available data, and no values were imputed. The point estimate for protocol completion met the predeclared 80% feasibility target, but its wide interval, the convenience sample, missing responses, fixed scenario order, self-selection, demand effects, absence of a control group, and lack of follow-up preclude causal or generalisable conclusions. The study-specific ordinal ratings describe perceived realism and usefulness only for the baseline workflow.

Completion was distributed across 14 workload, 13 boundary, and 12 relationship attempts. The mean paired confidence changes were +0.85, +0.85, and +1.25 for those scenarios, respectively, but the small, repeated-measures sample does not support scenario comparisons. Deterministic feedback also emitted lexical indicators for the 39 completed tasks; because these rules were not independently annotated or validated as measures of communication skill, they are implementation outputs rather than participant outcomes. Reporting them as success rates would overstate both the pilot and the extractor.

5.6 Threats to Validity

Model and statistical validity. IEMOCAP contains ten actors in acted or improvised English dialogue. Session-held-out folds prevent identity leakage for the tested speaker, but five dyads cannot represent demographic, cultural, linguistic, device, microphone, disability, or deployment diversity. Dialogue bootstrapping respects within-conversation dependence among the 151 observed dialogues but cannot create missing populations. One primary training seed leaves optimisation variance incompletely characterised, and small numerical variation can arise across GPU executions. Commit, configuration, prediction, and metric provenance is therefore necessary; provenance for private final model files remains incomplete.

The final full-data packages have no independent test set, and their pooled packaging diagnostics reuse labels across validation and reporting roles. They must not replace the held-out fold evidence. ECE depends on binning and may deteriorate under distribution shift; even favourable IEMOCAP calibration does not validate confidence in a deployment population. The 0.55 and 0.75 policy bands are operational presentation settings rather than psychological boundaries.

Construct and external validity. IEMOCAP labels are annotations of expressive categories, not direct observations of internal emotion. Merging excitement with happiness and mapping rare fear cases to exploratory anxiety simplify already contested constructs. Scores must not be interpreted as diagnosis, emotional truth,

or equitable performance. The six-class anxiety result is especially fragile because it contains only 40 mapped examples.

Application features are operational constructs as well. Lexical rules can locate phrases such as explicit requests or refusals in authored cases, but they do not measure general communication ability, intent, cause, readiness, resistance, cognition, or transfer to a real conversation. No clinician, validated questionnaire, independent annotation study, or held-out corpus establishes the cognitive fields as psychological measures. Deterministic feedback improves auditability, not construct validity. Negation, quotation, indirect language, dialect, culture, and multiple simultaneous requests remain likely failure sources.

Application and deployment validity. Engineering tests use controlled distributions, synthetic accounts, deterministic templates, and selected browser states. They do not establish trained-model latency, real-microphone robustness, live-provider variability, concurrent capacity, production security, backup operations, long-duration reliability, or absence of harm. Zero automated accessibility violations is narrower than WCAG conformance and does not substitute for screen-reader or participatory testing. The in-process benchmark excludes the components most sensitive to deployment infrastructure. A controlled deployment therefore requires target-population model validation, model and provider integration tests, staged load testing, recovery exercises, monitoring, and independent security and accessibility review.

Pilot validity. The pilot is small, single-arm, uncontrolled, incomplete in places, and based on study-specific ordinal measures. Most importantly, required tasks bypassed the multimodal and affect-aware path, so the pilot supplies no human evidence for the dissertation's adaptation mechanism. Immediate confidence change may reflect repetition, demand characteristics, or selection effects and cannot establish sustained learning or real-world transfer. The pilot is therefore retained only as contextual evidence that some participants completed and positively rated the baseline workflow.

In summary, the evaluation supports four bounded conclusions: causal context improved four-class text classification on session-held-out IEMOCAP; audio added complementary benchmark information but introduced a calibration trade-off under equal fusion; deterministic policies implemented the declared uncertainty, preference, fallback, and safety precedence; and the integrated offline core passed its scoped verification. Chapter 6 draws the final research conclusions without extending them beyond these boundaries.

Chapter 6

Conclusions and Future Work

6.1 Conclusions for the Research Questions

This dissertation asked whether conversational context and speech provide useful affective evidence, how such evidence can be used without giving a classifier authority over a role-play, and whether the resulting application meets its declared engineering requirements. The evaluation in Chapter 5 supports the following bounded answers.

RQ1: conversational context. Preceding dialogue turns improved text-based emotion classification under the declared four-class, speaker-independent IEMOCAP protocol. The paired dialogue-level comparison consistently favoured the contextual model, so the improvement cannot be attributed to different speakers, label sets, or optimisation settings. This is evidence that causal conversational context helps classify annotated expression in this benchmark; it is not evidence that the model recovers a person's private emotional state or that the gain will transfer unchanged to spontaneous AffectLab conversations. The exploratory six-class result points in the same aggregate direction but is too sparse to support an anxiety-recognition claim.

RQ2: multimodal complementarity and calibration. Acoustic information was complementary to contextual text even though audio was the weaker individual modality. Predeclared equal-weight fusion improved the principal classification measures over either unimodal model, with the paired bootstrap comparison favouring fusion. However, this configuration was substantially less calibrated than contextual text. Validation-fitted fusion recovered strong calibration while retaining most of the classification gain. The answer is therefore conditional: multimodal fusion improved benchmark classification, but the configuration that maximised classification was not the configuration with the most trustworthy probabilities. Classification and calibration must be selected and reported separately when confidence affects adaptive behaviour.

RQ3: bounded use of affective evidence. Uncertain predictions can be integrated responsibly by restricting what they are allowed to change. AffectLab records the prediction and its provenance, checks confidence, modality agreement,

availability, eligibility, and user preference, and maps the result to a small set of reversible pacing actions. Deterministic scenario controllers remain authoritative for progression and outcomes; observable language evidence determines feedback; explicit safety handling takes precedence; and inference failure preserves the baseline workflow. Authored policy cases and integrated journeys conformed to these rules. This establishes explainable and failure-tolerant policy behaviour under tested conditions, not that an affect estimate is correct for a particular user or that adaptation improves rehearsal outcomes.

RQ4: integrated application quality. The completed application satisfied its scoped functional and non-functional requirements for the tested offline deterministic core. Backend, production-frontend, browser, automated-accessibility, persistence, recovery, and full-stack checks supported the declared behaviours, while the local regression benchmark met every response-time budget. These layers establish an implementation-complete, deployment-oriented system rather than deployment readiness. The performance run excluded real network transit, production persistence, trained inference, external providers, concurrency, and adversarial traffic. A controlled deployment therefore remains conditional on target-environment performance, security, observability, backup recovery, and incident-response validation.

6.2 Contributions and Material Limitations

The work makes three connected contributions. Empirically, it provides a controlled speaker-independent comparison of utterance-only and contextual text classification, followed by a text–audio study that treats calibration as distinct from classification. Architecturally, it demonstrates a hybrid role-play system in which probabilistic models and optional generation influence presentation without controlling dialogue outcomes, assessment, or safety. Methodologically, it separates benchmark evidence, deterministic policy conformance, application verification, local performance, and supplementary user feedback so that one evidence type is not used to support a stronger claim than it can bear. Versioned policies, provenance, evidence-grounded feedback, fallbacks, and replayable state make the implemented decisions inspectable.

These contributions are bounded by the available data and evaluation. IEMOCAP contains ten actors in acted or improvised English dialogue and does not provide external, cultural, device, or deployment validation. A single primary training seed leaves optimisation variability incompletely characterised, while full-data packages and the operational fusion summary have no independent held-out estimate. Emotion categories and confidence thresholds are operational signals rather than psychological measurements. The three scenario policies and lexical feedback rules cover limited forms of communication, and the separately implemented cognitive support-routing fields are unvalidated heuristics, not measures of cognition, intention, readiness, resistance, or clinical state.

The engineering tests used controlled inputs and synthetic accounts, and automated accessibility checks do not replace evaluation with assistive-technology users. Most importantly, the small, uncontrolled pilot exercised the deterministic baseline rather than multimodal inference or affect-aware pacing. It can inform workflow

refinement but cannot establish adaptation benefit, sustained learning, transfer, or absence of harm. AffectLab is consequently a communication-practice research application, not a diagnostic, clinical, crisis, or therapeutic system.

6.3 Prioritised Future Work

The immediate priority is deployment-specific validation of the existing system. A staging study should exercise the packaged models, external providers, intended database, real network path, and representative concurrency. Declared service objectives, load and soak tests, provider and database failures, backup restoration, security review, monitoring, and incident exercises should establish latency, capacity, recovery, and operational risk before any production-readiness claim.

The next priority is to strengthen the evidence behind both inference and interaction. Cross-corpus and target-context evaluation should examine calibration under language, microphone, noise, transcription, and demographic variation, using repeated seeds and prospective held-out data. Independent human annotation should assess the validity and error patterns of feedback rules; psychologically suggestive support-routing labels should be removed, renamed as observable behaviour, or subjected to appropriate expert and ethics-led validation. A preregistered comparison of the baseline and bounded affect-aware policy should then measure helpfulness, intrusiveness, task outcomes, and user preference without treating engagement as benefit.

Longer-term work should use participatory accessibility testing, multilingual and cultural adaptation, and longitudinal studies of retention and transfer. These studies should preserve opt-in controls, data minimisation, transparent uncertainty, and a useful non-adaptive baseline. Model, prompt, scenario, threshold, and safety-resource changes should remain versioned and governed as the evidence base develops.

6.4 Final Conclusion

This dissertation shows that contextual and acoustic affective evidence can improve benchmark classification and can be connected to role-play without making emotion recognition authoritative. AffectLab combines bounded adaptation with deterministic progression, evidence-grounded feedback, safety precedence, user control, and auditable fallbacks. Its evaluation establishes the implemented system and policies within clearly defined test conditions, while leaving claims about emotional truth, human benefit, and production readiness to future evidence. Within those limits, the work provides a reproducible foundation for studying emotion-aware communication rehearsal while preserving uncertainty and user agency.

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Appendix A

Generative AI Tools Usage

Generative AI tools were used as supplementary resources during the development and documentation of this thesis. Their use was limited to supporting literature exploration, improving the clarity of academic writing, assisting with LaTeX formatting, and providing guidance during technical problem-solving. All generated suggestions were reviewed before being incorporated into the thesis.

An overview of the generative AI tools used is presented in Table A.1.

AI Tool	Use Case	Used in	Remarks
ChatGPT 5.6 Sol / 6 Astra	Literature exploration and identification of relevant scientific publications	Chapter 2	Suggested references were independently verified
	Improvement of academic phrasing, grammar, and text clarity	Chapters 1–6	Suggestions were reviewed and adapted
	Assistance with LaTeX syntax, formatting, and document structure	Entire thesis	Formatting suggestions were manually checked
	Technical guidance and debugging suggestions	Implementation chapters	Suggestions were evaluated and tested before use

Table A.1: Overview of generative AI usage

A.1 ChatGPT Details

ChatGPT (versions 5.6 Sol and 6 Astra, developed by OpenAI) was used as a supplementary assistant throughout selected stages of the thesis. Its contributions are described below.

A.1.1 Literature Exploration

Purpose: To support the identification of relevant scientific publications and research directions related to the topics investigated in this thesis.

Method: ChatGPT was prompted to suggest academic publications, relevant research topics, and potential directions for organizing the state-of-the-art review in Chapter 2.

Verification: Suggested publications were checked against academic sources to verify their existence, relevance, and bibliographic information. The original publications were consulted before being referenced.

Implementation: The verified references and research directions were used as supporting resources when developing the literature review.

A.1.2 Academic Writing Assistance

Purpose: To improve the readability, grammatical correctness, and academic style of selected passages.

Method: Selected paragraphs were provided to ChatGPT with requests to improve sentence structure, clarity, or academic phrasing while preserving their original meaning.

Verification: The suggested revisions were reviewed to ensure that the technical meaning and intended interpretation remained unchanged.

Implementation: Selected linguistic improvements were incorporated into the thesis after manual revision.

A.1.3 LaTeX Formatting Assistance

Purpose: To assist with document formatting and the resolution of LaTeX-related issues.

Method: ChatGPT was consulted for guidance on LaTeX commands, table formatting, document structure, and compilation errors.

Verification: Suggested solutions were checked by compiling the document and inspecting the resulting output.

Implementation: Validated formatting solutions were applied where appropriate.

A.1.4 Technical Guidance and Debugging

Purpose: To obtain supplementary explanations and troubleshooting suggestions during the implementation process.

Method: ChatGPT was occasionally consulted to clarify technical concepts, explain error messages, and suggest possible approaches to resolving implementation difficulties.

Verification: Suggestions were assessed against the project requirements and, where applicable, tested in the development environment.

Implementation: Only solutions considered appropriate and successfully validated were incorporated into the implementation.

Generative AI was used exclusively as a supporting tool. The selection of research methods, system design decisions, implementation validation, interpretation

of experimental results, and final conclusions remained the responsibility of the author.